



Name: _____

Date: _____

Score: / 30

Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Euler method

1. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 1x - 4$, $y(2) = 4$.

$$y' = 1x - 4, x_0 = 2, y_0 = 4, h = 1$$

Answer: _____

2. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 5$ starting at $(x_0, y_0) = (2, 8)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 5, x_0 = 2, y_0 = 8, h = 1$$

Answer: _____

3. A population model satisfies $y' = 4t + 10$. Starting at $t = 0$, $y = 10$, use one step of Euler's method with $h = 2$ to estimate $y(0 + 2)$.

$$y' = 4x + 10, x_0 = 0, y_0 = 10, h = 2$$

Answer: _____

4. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 3x + 2$, $y(3) = 5$.

$$y' = 3x + 2, x_0 = 3, y_0 = 5, h = 1$$

Answer: _____

5. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 7$ starting at $(x_0, y_0) = (0, 6)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 7, x_0 = 0, y_0 = 6, h = 1$$

Answer: _____



6. A population model satisfies $y' = 3t + 3$. Starting at $t = 3$, $y = 19$, use one step of Euler's method with $h = 2$ to estimate $y(3 + 2)$.

$$y' = 3x + 3, \quad x_0 = 3, \quad y_0 = 19, \quad h = 2$$

Answer: _____

7. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 5x + 1$, $y(1) = 9$.

$$y' = 5x + 1, \quad x_0 = 1, \quad y_0 = 9, \quad h = 2$$

Answer: _____

8. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 6$ starting at $(x_0, y_0) = (1, 2)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 6, \quad x_0 = 1, \quad y_0 = 2, \quad h = 1$$

Answer: _____

9. A population model satisfies $y' = 4t + 6$. Starting at $t = 2$, $y = 12$, use one step of Euler's method with $h = 2$ to estimate $y(2 + 2)$.

$$y' = 4x + 6, \quad x_0 = 2, \quad y_0 = 12, \quad h = 2$$

Answer: _____

10. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 2x + 4$, $y(1) = 4$.

$$y' = 2x + 4, \quad x_0 = 1, \quad y_0 = 4, \quad h = 1$$

Answer: _____

11. Apply Euler's method (one step, $h = 1$) to $y' = 4x + 1$ starting at $(x_0, y_0) = (2, 6)$. What is the approximation at $x = x_0 + 1$?

$$y' = 4x + 1, \quad x_0 = 2, \quad y_0 = 6, \quad h = 1$$

Answer: _____

12. A population model satisfies $y' = 2t + 9$. Starting at $t = 3$, $y = 16$, use one step of Euler's method with $h = 2$ to estimate $y(3 + 2)$.

$$y' = 2x + 9, \quad x_0 = 3, \quad y_0 = 16, \quad h = 2$$

Answer: _____

13. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 2x + 3$, $y(2) = 8$.

$$y' = 2x + 3, \quad x_0 = 2, \quad y_0 = 8, \quad h = 2$$

Answer: _____



14. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 4$ starting at $(x_0, y_0) = (0, 7)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 4, \quad x_0 = 0, \quad y_0 = 7, \quad h = 1$$

Answer: _____

15. A population model satisfies $y' = 1t + 9$. Starting at $t = 2, y = 18$, use one step of Euler's method with $h = 1$ to estimate $y(2 + 1)$.

$$y' = 1x + 9, \quad x_0 = 2, \quad y_0 = 18, \quad h = 1$$

Answer: _____

16. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 1x + 1, y(0) = 4$.

$$y' = 1x + 1, \quad x_0 = 0, \quad y_0 = 4, \quad h = 2$$

Answer: _____

17. Apply Euler's method (one step, $h = 1$) to $y' = 6x + 4$ starting at $(x_0, y_0) = (0, 7)$. What is the approximation at $x = x_0 + 1$?

$$y' = 6x + 4, \quad x_0 = 0, \quad y_0 = 7, \quad h = 1$$

Answer: _____

18. A population model satisfies $y' = 2t + 3$. Starting at $t = 3, y = 5$, use one step of Euler's method with $h = 1$ to estimate $y(3 + 1)$.

$$y' = 2x + 3, \quad x_0 = 3, \quad y_0 = 5, \quad h = 1$$

Answer: _____

19. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 3x + 0, y(1) = 5$.

$$y' = 3x + 0, \quad x_0 = 1, \quad y_0 = 5, \quad h = 1$$

Answer: _____

20. Apply Euler's method (one step, $h = 1$) to $y' = 5x + 3$ starting at $(x_0, y_0) = (2, 3)$. What is the approximation at $x = x_0 + 1$?

$$y' = 5x + 3, \quad x_0 = 2, \quad y_0 = 3, \quad h = 1$$

Answer: _____

21. A population model satisfies $y' = 2t + 3$. Starting at $t = 1, y = 19$, use one step of Euler's method with $h = 2$ to estimate $y(1 + 2)$.

$$y' = 2x + 3, \quad x_0 = 1, \quad y_0 = 19, \quad h = 2$$

Answer: _____



22. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 4x + 3$, $y(2) = 5$.

$$y' = 4x + 3, \quad x_0 = 2, \quad y_0 = 5, \quad h = 2$$

Answer: _____

23. Apply Euler's method (one step, $h = 1$) to $y' = 5x + 2$ starting at $(x_0, y_0) = (2, 6)$. What is the approximation at $x = x_0 + 1$?

$$y' = 5x + 2, \quad x_0 = 2, \quad y_0 = 6, \quad h = 1$$

Answer: _____

24. A population model satisfies $y' = 4t + 6$. Starting at $t = 2$, $y = 12$, use one step of Euler's method with $h = 2$ to estimate $y(2 + 2)$.

$$y' = 4x + 6, \quad x_0 = 2, \quad y_0 = 12, \quad h = 2$$

Answer: _____

25. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 1x + 0$, $y(3) = 4$.

$$y' = 1x + 0, \quad x_0 = 3, \quad y_0 = 4, \quad h = 2$$

Answer: _____

26. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 2$ starting at $(x_0, y_0) = (1, 5)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 2, \quad x_0 = 1, \quad y_0 = 5, \quad h = 1$$

Answer: _____

27. A population model satisfies $y' = 4t + 10$. Starting at $t = 0$, $y = 12$, use one step of Euler's method with $h = 2$ to estimate $y(0 + 2)$.

$$y' = 4x + 10, \quad x_0 = 0, \quad y_0 = 12, \quad h = 2$$

Answer: _____

28. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 1x + -3$, $y(1) = 2$.

$$y' = 1x - 3, \quad x_0 = 1, \quad y_0 = 2, \quad h = 2$$

Answer: _____

29. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 5$ starting at $(x_0, y_0) = (2, 5)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 5, \quad x_0 = 2, \quad y_0 = 5, \quad h = 1$$

Answer: _____



30. A population model satisfies $y' = 3t + 10$. Starting at $t = 0$, $y = 13$, use one step of Euler's method with $h = 1$ to estimate $y(0 + 1)$.

$$y' = 3x + 10, \quad x_0 = 0, \quad y_0 = 13, \quad h = 1$$

Answer: _____





MATH240: Numerical Methods

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ANSWER KEY & SOLUTIONS

Topics: Euler method. All answers verified by independent computation.

Solutions



Euler method

1. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 1x - 4$, $y(2) = 4$.

$$y' = 1x - 4, \quad x_0 = 2, \quad y_0 = 4, \quad h = 1$$

$$\rightarrow f(x_0) = 1 \cdot 2 - 4 = -2.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 4 + 1 \cdot (-2) = 2.$$

Answer: $y_1 \approx 4 + 1 \cdot -2 = 2$

2. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 5$ starting at $(x_0, y_0) = (2, 8)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 5, \quad x_0 = 2, \quad y_0 = 8, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(2) = 2 \cdot 2 + 5 = 9.$$

$$\rightarrow \text{Next approximation: } y_1 = 8 + 1 \cdot 9 = 17.$$

Answer: $y_1 \approx 8 + 1 \cdot 9 = 17$

3. A population model satisfies $y' = 4t + 10$. Starting at $t = 0$, $y = 10$, use one step of Euler's method with $h = 2$ to estimate $y(0 + 2)$.

$$y' = 4x + 10, \quad x_0 = 0, \quad y_0 = 10, \quad h = 2$$

$$\rightarrow \text{Rate at } t = 0: f(0) = 4 \cdot 0 + 10 = 10.$$

$$\rightarrow \text{Euler step: } y_1 = 10 + 2 \cdot 10 = 30.$$

Answer: $y_1 \approx 10 + 2 \cdot 10 = 30$

4. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 3x + 2$, $y(3) = 5$.

$$y' = 3x + 2, \quad x_0 = 3, \quad y_0 = 5, \quad h = 1$$

$$\rightarrow f(x_0) = 3 \cdot 3 + 2 = 11.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 5 + 1 \cdot 11 = 16.$$

Answer: $y_1 \approx 5 + 1 \cdot 11 = 16$

5. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 7$ starting at $(x_0, y_0) = (0, 6)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 7, \quad x_0 = 0, \quad y_0 = 6, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(0) = 2 \cdot 0 + 7 = 7.$$

$$\rightarrow \text{Next approximation: } y_1 = 6 + 1 \cdot 7 = 13.$$

Answer: $y_1 \approx 6 + 1 \cdot 7 = 13$

6. A population model satisfies $y' = 3t + 3$. Starting at $t = 3$, $y = 19$, use one step of Euler's method with $h = 2$ to estimate $y(3 + 2)$.

$$y' = 3x + 3, \quad x_0 = 3, \quad y_0 = 19, \quad h = 2$$

$$\rightarrow \text{Rate at } t = 3: f(3) = 3 \cdot 3 + 3 = 12.$$

$$\rightarrow \text{Euler step: } y_1 = 19 + 2 \cdot 12 = 43.$$

Answer: $y_1 \approx 19 + 2 \cdot 12 = 43$



7. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 5x + 1$, $y(1) = 9$.

$$y' = 5x + 1, \quad x_0 = 1, \quad y_0 = 9, \quad h = 2$$

$$\rightarrow f(x_0) = 5 \cdot 1 + 1 = 6.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 9 + 2 \cdot 6 = 21.$$

Answer: $y_1 \approx 9 + 2 \cdot 6 = 21$

8. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 6$ starting at $(x_0, y_0) = (1, 2)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 6, \quad x_0 = 1, \quad y_0 = 2, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(1) = 2 \cdot 1 + 6 = 8.$$

$$\rightarrow \text{Next approximation: } y_1 = 2 + 1 \cdot 8 = 10.$$

Answer: $y_1 \approx 2 + 1 \cdot 8 = 10$

9. A population model satisfies $y' = 4t + 6$. Starting at $t = 2$, $y = 12$, use one step of Euler's method with $h = 2$ to estimate $y(2 + 2)$.

$$y' = 4x + 6, \quad x_0 = 2, \quad y_0 = 12, \quad h = 2$$

$$\rightarrow \text{Rate at } t = 2: f(2) = 4 \cdot 2 + 6 = 14.$$

$$\rightarrow \text{Euler step: } y_1 = 12 + 2 \cdot 14 = 40.$$

Answer: $y_1 \approx 12 + 2 \cdot 14 = 40$

10. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 2x + 4$, $y(1) = 4$.

$$y' = 2x + 4, \quad x_0 = 1, \quad y_0 = 4, \quad h = 1$$

$$\rightarrow f(x_0) = 2 \cdot 1 + 4 = 6.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 4 + 1 \cdot 6 = 10.$$

Answer: $y_1 \approx 4 + 1 \cdot 6 = 10$

11. Apply Euler's method (one step, $h = 1$) to $y' = 4x + 1$ starting at $(x_0, y_0) = (2, 6)$. What is the approximation at $x = x_0 + 1$?

$$y' = 4x + 1, \quad x_0 = 2, \quad y_0 = 6, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(2) = 4 \cdot 2 + 1 = 9.$$

$$\rightarrow \text{Next approximation: } y_1 = 6 + 1 \cdot 9 = 15.$$

Answer: $y_1 \approx 6 + 1 \cdot 9 = 15$

12. A population model satisfies $y' = 2t + 9$. Starting at $t = 3$, $y = 16$, use one step of Euler's method with $h = 2$ to estimate $y(3 + 2)$.

$$y' = 2x + 9, \quad x_0 = 3, \quad y_0 = 16, \quad h = 2$$

$$\rightarrow \text{Rate at } t = 3: f(3) = 2 \cdot 3 + 9 = 15.$$

$$\rightarrow \text{Euler step: } y_1 = 16 + 2 \cdot 15 = 46.$$

Answer: $y_1 \approx 16 + 2 \cdot 15 = 46$



13. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 2x + 3$, $y(2) = 8$.

$$y' = 2x + 3, \quad x_0 = 2, \quad y_0 = 8, \quad h = 2$$

$$\rightarrow f(x_0) = 2 \cdot 2 + 3 = 7.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 8 + 2 \cdot 7 = 22.$$

Answer: $y_1 \approx 8 + 2 \cdot 7 = 22$

14. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 4$ starting at $(x_0, y_0) = (0, 7)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 4, \quad x_0 = 0, \quad y_0 = 7, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(0) = 2 \cdot 0 + 4 = 4.$$

$$\rightarrow \text{Next approximation: } y_1 = 7 + 1 \cdot 4 = 11.$$

Answer: $y_1 \approx 7 + 1 \cdot 4 = 11$

15. A population model satisfies $y' = 1t + 9$. Starting at $t = 2$, $y = 18$, use one step of Euler's method with $h = 1$ to estimate $y(2 + 1)$.

$$y' = 1x + 9, \quad x_0 = 2, \quad y_0 = 18, \quad h = 1$$

$$\rightarrow \text{Rate at } t = 2: f(2) = 1 \cdot 2 + 9 = 11.$$

$$\rightarrow \text{Euler step: } y_1 = 18 + 1 \cdot 11 = 29.$$

Answer: $y_1 \approx 18 + 1 \cdot 11 = 29$

16. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 1x + 1$, $y(0) = 4$.

$$y' = 1x + 1, \quad x_0 = 0, \quad y_0 = 4, \quad h = 2$$

$$\rightarrow f(x_0) = 1 \cdot 0 + 1 = 1.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 4 + 2 \cdot 1 = 6.$$

Answer: $y_1 \approx 4 + 2 \cdot 1 = 6$

17. Apply Euler's method (one step, $h = 1$) to $y' = 6x + 4$ starting at $(x_0, y_0) = (0, 7)$. What is the approximation at $x = x_0 + 1$?

$$y' = 6x + 4, \quad x_0 = 0, \quad y_0 = 7, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(0) = 6 \cdot 0 + 4 = 4.$$

$$\rightarrow \text{Next approximation: } y_1 = 7 + 1 \cdot 4 = 11.$$

Answer: $y_1 \approx 7 + 1 \cdot 4 = 11$

18. A population model satisfies $y' = 2t + 3$. Starting at $t = 3$, $y = 5$, use one step of Euler's method with $h = 1$ to estimate $y(3 + 1)$.

$$y' = 2x + 3, \quad x_0 = 3, \quad y_0 = 5, \quad h = 1$$

$$\rightarrow \text{Rate at } t = 3: f(3) = 2 \cdot 3 + 3 = 9.$$

$$\rightarrow \text{Euler step: } y_1 = 5 + 1 \cdot 9 = 14.$$

Answer: $y_1 \approx 5 + 1 \cdot 9 = 14$



19. Use Euler's method with step size $h = 1$ to approximate $y(x_0 + h)$ for $y' = 3x + 0$, $y(1) = 5$.

$$y' = 3x + 0, \quad x_0 = 1, \quad y_0 = 5, \quad h = 1$$

$$\rightarrow f(x_0) = 3 \cdot 1 + 0 = 3.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 5 + 1 \cdot 3 = 8.$$

Answer: $y_1 \approx 5 + 1 \cdot 3 = 8$

20. Apply Euler's method (one step, $h = 1$) to $y' = 5x + 3$ starting at $(x_0, y_0) = (2, 3)$. What is the approximation at $x = x_0 + 1$?

$$y' = 5x + 3, \quad x_0 = 2, \quad y_0 = 3, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(2) = 5 \cdot 2 + 3 = 13.$$

$$\rightarrow \text{Next approximation: } y_1 = 3 + 1 \cdot 13 = 16.$$

Answer: $y_1 \approx 3 + 1 \cdot 13 = 16$

21. A population model satisfies $y' = 2t + 3$. Starting at $t = 1$, $y = 19$, use one step of Euler's method with $h = 2$ to estimate $y(1 + 2)$.

$$y' = 2x + 3, \quad x_0 = 1, \quad y_0 = 19, \quad h = 2$$

$$\rightarrow \text{Rate at } t = 1: f(1) = 2 \cdot 1 + 3 = 5.$$

$$\rightarrow \text{Euler step: } y_1 = 19 + 2 \cdot 5 = 29.$$

Answer: $y_1 \approx 19 + 2 \cdot 5 = 29$

22. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 4x + 3$, $y(2) = 5$.

$$y' = 4x + 3, \quad x_0 = 2, \quad y_0 = 5, \quad h = 2$$

$$\rightarrow f(x_0) = 4 \cdot 2 + 3 = 11.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 5 + 2 \cdot 11 = 27.$$

Answer: $y_1 \approx 5 + 2 \cdot 11 = 27$

23. Apply Euler's method (one step, $h = 1$) to $y' = 5x + 2$ starting at $(x_0, y_0) = (2, 6)$. What is the approximation at $x = x_0 + 1$?

$$y' = 5x + 2, \quad x_0 = 2, \quad y_0 = 6, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(2) = 5 \cdot 2 + 2 = 12.$$

$$\rightarrow \text{Next approximation: } y_1 = 6 + 1 \cdot 12 = 18.$$

Answer: $y_1 \approx 6 + 1 \cdot 12 = 18$

24. A population model satisfies $y' = 4t + 6$. Starting at $t = 2$, $y = 12$, use one step of Euler's method with $h = 2$ to estimate $y(2 + 2)$.

$$y' = 4x + 6, \quad x_0 = 2, \quad y_0 = 12, \quad h = 2$$

$$\rightarrow \text{Rate at } t = 2: f(2) = 4 \cdot 2 + 6 = 14.$$

$$\rightarrow \text{Euler step: } y_1 = 12 + 2 \cdot 14 = 40.$$

Answer: $y_1 \approx 12 + 2 \cdot 14 = 40$



25. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 1x + 0$, $y(3) = 4$.

$$y' = 1x + 0, \quad x_0 = 3, \quad y_0 = 4, \quad h = 2$$

$$\rightarrow f(x_0) = 1 \cdot 3 + 0 = 3.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 4 + 2 \cdot 3 = 10.$$

Answer: $y_1 \approx 4 + 2 \cdot 3 = 10$

26. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 2$ starting at $(x_0, y_0) = (1, 5)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 2, \quad x_0 = 1, \quad y_0 = 5, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(1) = 2 \cdot 1 + 2 = 4.$$

$$\rightarrow \text{Next approximation: } y_1 = 5 + 1 \cdot 4 = 9.$$

Answer: $y_1 \approx 5 + 1 \cdot 4 = 9$

27. A population model satisfies $y' = 4t + 10$. Starting at $t = 0$, $y = 12$, use one step of Euler's method with $h = 2$ to estimate $y(0 + 2)$.

$$y' = 4x + 10, \quad x_0 = 0, \quad y_0 = 12, \quad h = 2$$

$$\rightarrow \text{Rate at } t = 0: f(0) = 4 \cdot 0 + 10 = 10.$$

$$\rightarrow \text{Euler step: } y_1 = 12 + 2 \cdot 10 = 32.$$

Answer: $y_1 \approx 12 + 2 \cdot 10 = 32$

28. Use Euler's method with step size $h = 2$ to approximate $y(x_0 + h)$ for $y' = 1x - 3$, $y(1) = 2$.

$$y' = 1x - 3, \quad x_0 = 1, \quad y_0 = 2, \quad h = 2$$

$$\rightarrow f(x_0) = 1 \cdot 1 - 3 = -2.$$

$$\rightarrow y_1 = y_0 + h \cdot f(x_0) = 2 + 2 \cdot -2 = -2.$$

Answer: $y_1 \approx 2 + 2 \cdot -2 = -2$

29. Apply Euler's method (one step, $h = 1$) to $y' = 2x + 5$ starting at $(x_0, y_0) = (2, 5)$. What is the approximation at $x = x_0 + 1$?

$$y' = 2x + 5, \quad x_0 = 2, \quad y_0 = 5, \quad h = 1$$

$$\rightarrow \text{Evaluate slope: } f(2) = 2 \cdot 2 + 5 = 9.$$

$$\rightarrow \text{Next approximation: } y_1 = 5 + 1 \cdot 9 = 14.$$

Answer: $y_1 \approx 5 + 1 \cdot 9 = 14$

30. A population model satisfies $y' = 3t + 10$. Starting at $t = 0$, $y = 13$, use one step of Euler's method with $h = 1$ to estimate $y(0 + 1)$.

$$y' = 3x + 10, \quad x_0 = 0, \quad y_0 = 13, \quad h = 1$$

$$\rightarrow \text{Rate at } t = 0: f(0) = 3 \cdot 0 + 10 = 10.$$

$$\rightarrow \text{Euler step: } y_1 = 13 + 1 \cdot 10 = 23.$$

Answer: $y_1 \approx 13 + 1 \cdot 10 = 23$

